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Chapter 1

Introduction to Computer Intelligence and Machine Learning

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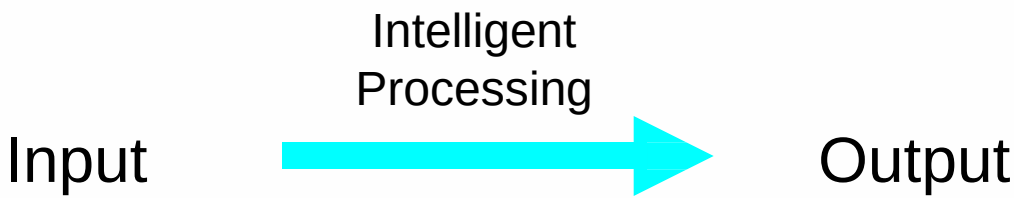
Intelligence

- Intelligence
  - Ability to solve problem correctly
    - Examples of Intelligent Behaviors or Tasks
  - Classification of texts based on content
  - Heart disease diagnosis
  - Chess playing

Artificial Intelligence

- Artificial Intelligence
  - Ability of machines in conducting intelligent tasks

- Intelligent Programs
  - Programs conducting specific intelligent tasks



Artificial Intelligence

A program that:

1)Acts like human (Turing test)

2)Thinks like human (human-like patterns of thinking steps)

3)Acts rationally (correctly)

4)Thinks rationally (logically)

Computer Intelligence

- Definitions (Russel et al, 1998)
  - A methodology involving computing that exhibits an ability to learn and/or to deal with new situations, such that the system is perceived to possess one or more attributes of reason, such as generalization, discovery, association and abstraction (Russel et al, 1998).

CI Techniques

Genetic Algorithm
Neural Network
Fuzzy Logic
Support Vector Machine
K-mean
Hybrid Algorithm

CI Applications

Searching
Forecasting
Classification
Scheduling
Visualization
Modelling and Simulation
Pattern Recognition
Data Mining

Machine Learning

Machine Learning

Machine = Robots, Car, Airplane, Computer..
Etc

Our class:

Machine used = Computer

Computer Learning = Computer Intelligence

Machine Learning

- Machine Learning (Mitchell 1997)
  - Learn from past experiences
  - Improve the performances of intelligent programs
- Definitions (Mitchell 1997)
  - A computer program is said to learn from experience *E* with respect to some class of tasks *T* and performance measure *P*, if its performance at the tasks improves with the experiences

About“Learning”

- Learning general models from a data of particular examples
- Data is cheap and abundant (data warehouses, data marts); knowledge is expensive and scarce.
- Example in retail: Customer transactions to consumer behavior:

  People who bought “Da Vinci Code” also bought “The Five People You Meet in Heaven” (www.amazon.com)

- Build a model that is *a good and useful approximation* to the data.

Applications of Machine Learning

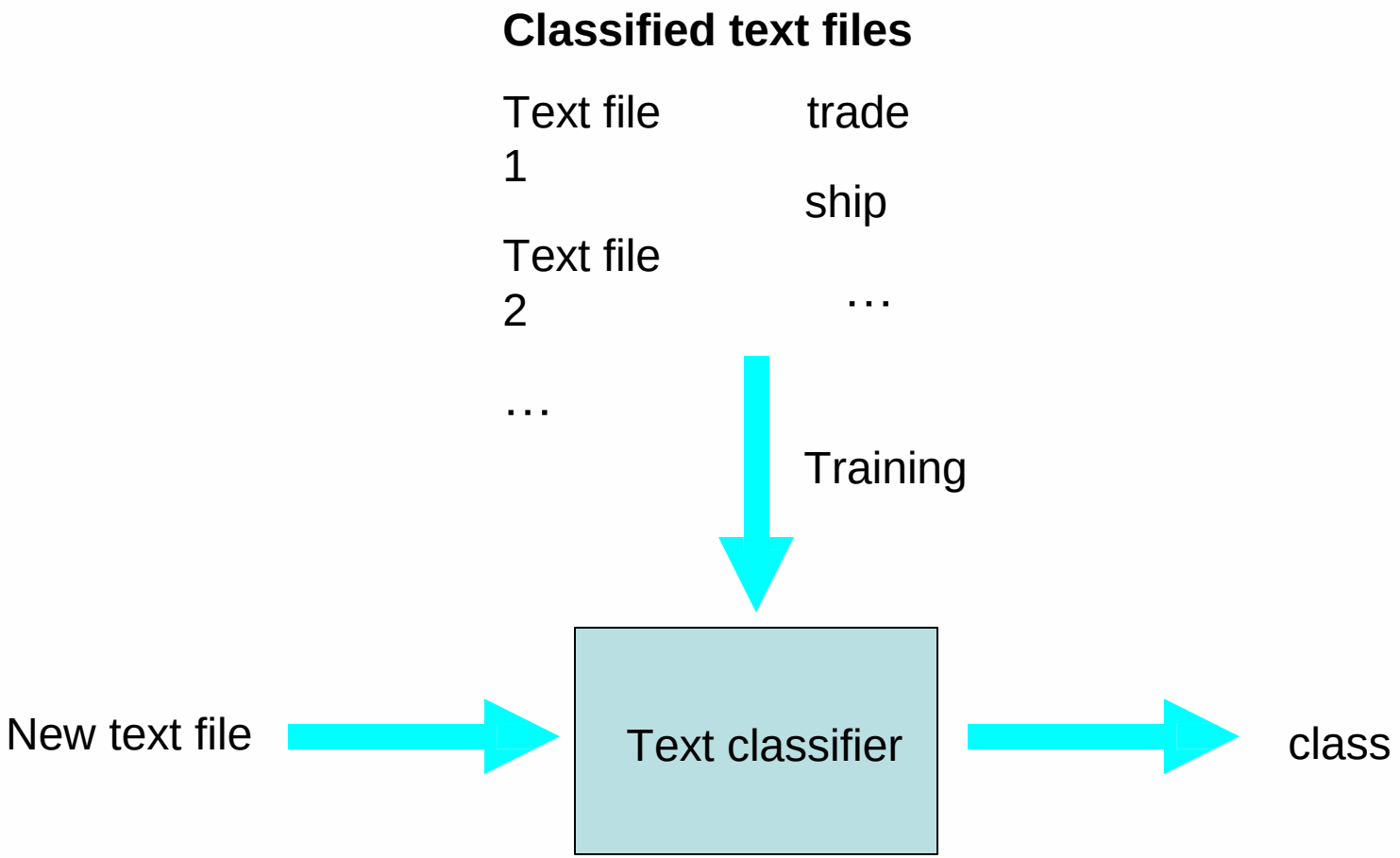
- Retail: Market basket analysis, Customer relationship management (CRM)
- Finance: Credit scoring, fraud detection
- Manufacturing: Optimization, troubleshooting
- Medicine: Medical diagnosis
- Telecommunications: Quality of service optimization
- Bioinformatics: Motifs, alignment
- Web mining: Search engines
- ...

Why is Machine Learning Important?

- Some tasks cannot be defined well, except by examples (e.g., recognizing people).
- Relationships and correlations can be hidden within large amounts of data. Machine Learning/Data Mining may be able to find these relationships.
- Human designers often produce machines that do not work as well as desired in the environments in which they are used.

- The amount of knowledge available about certain tasks might be too large for explicit encoding by humans (e.g., medical diagnostic).
- Environments change over time.
- New knowledge about tasks is constantly being discovered by humans. It may be difficult to continuously re-design systems “by hand”.

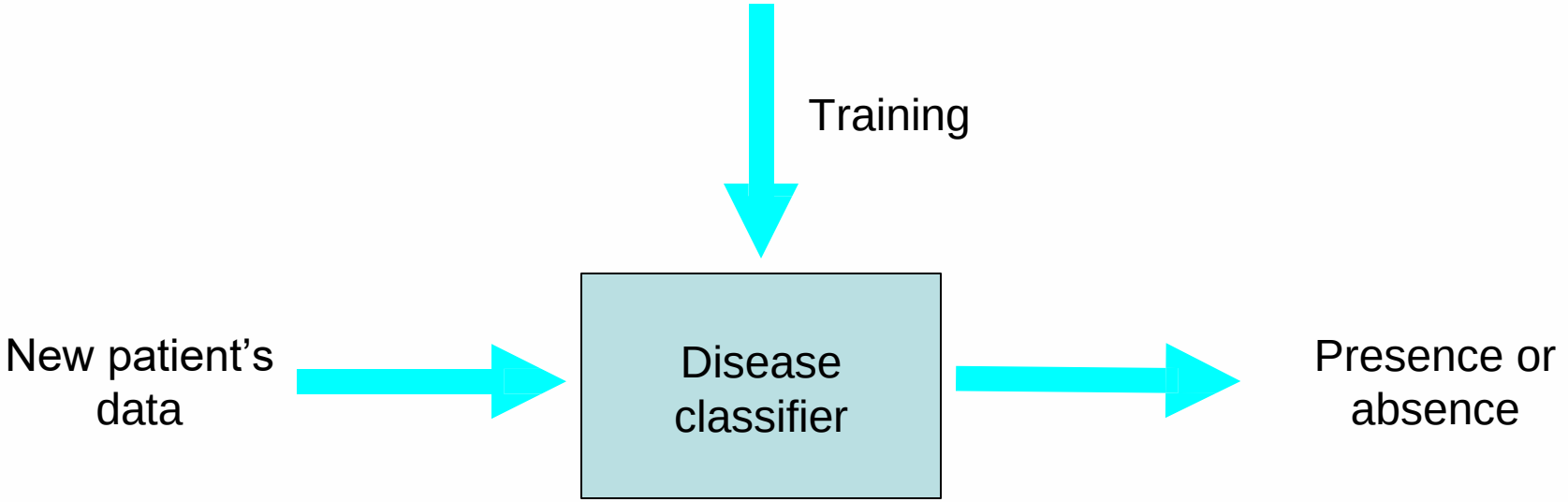
Example 1: Text Classification



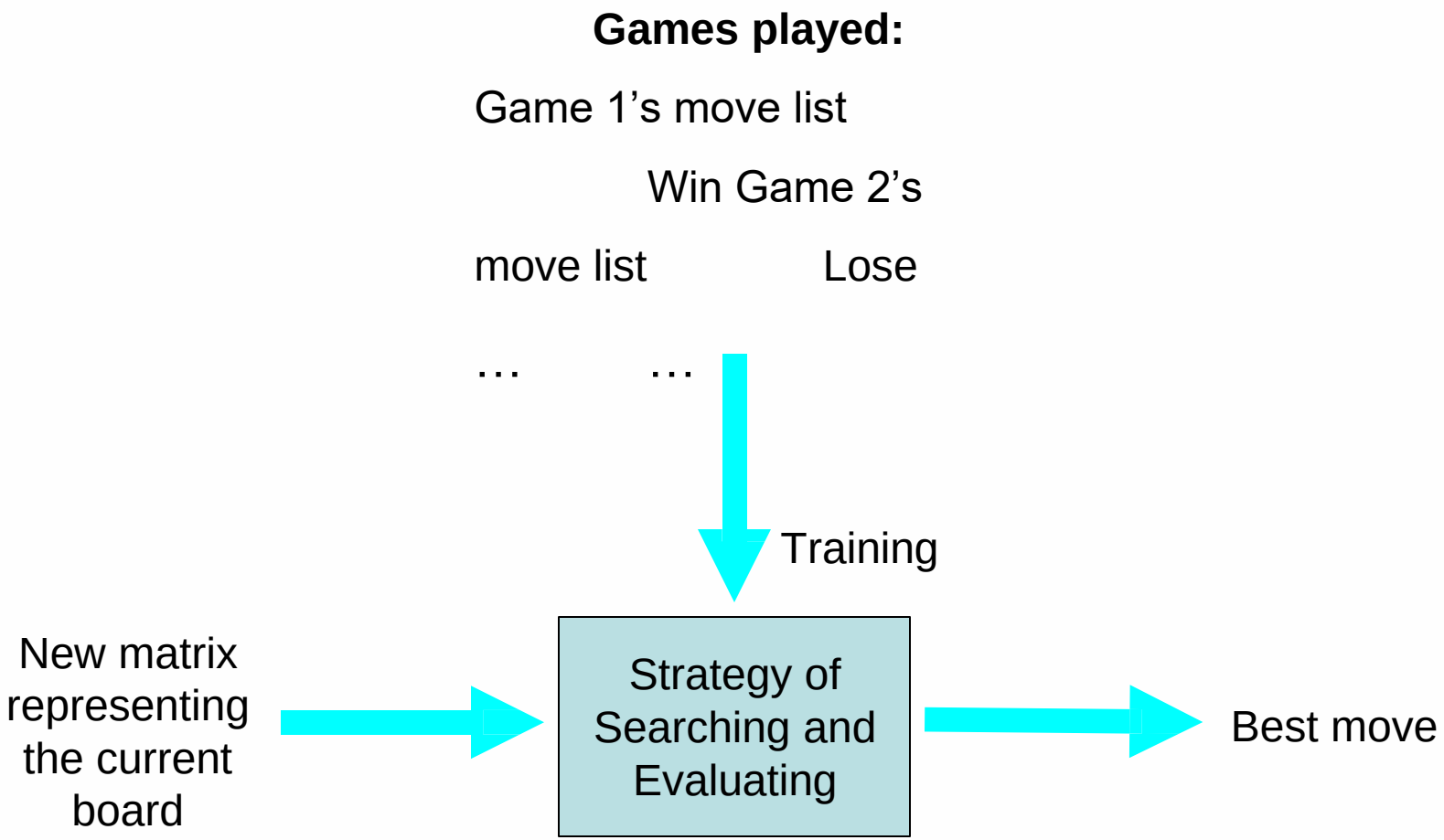
Example 2: Disease Diagnosis

Database of medical records

Patient 1's data	Absence
Patient 2's data	Presence
…	…



Example 3: Chess Playing



Areas of Influence for Machine Learning

- **Statistics:** How best to use samples drawn from unknown probability distributions to help decide from which distribution some new sample is drawn?
- **Brain Models:** Non-linear elements with weighted inputs (Artificial Neural Networks) have been suggested as simple models of biological neurons.
- **Adaptive Control Theory:** How to deal with controlling a process having unknown parameters that must be estimated during operation?

- **Psychology:** How to model human performance on various learning tasks?
- **Artificial Intelligence:** How to write algorithms to acquire the knowledge humans are able to acquire, at least, as well as humans?
- **Evolutionary Models:** How to model certain aspects of biological evolution to improve the performance of computer programs?

Why Machine Learning Is Possible?

- Mass Storage
  - More data available
- Higher Performance of Computer
  - Larger memory in handling the data
  - Greater computational power for calculating and even online learning

Advantages

- Alleviate Knowledge Acquisition Bottleneck
  - Does not require knowledge engineers
  - Scalable in constructing knowledge base
- Adaptive
  - Adaptive to the changing conditions
  - Easy in migrating to new domains

ML in a Nutshell

- Tens of thousands of machine learning algorithms
- Hundreds new every year
- Most machine learning algorithm has three components:
  - **Representation**
  - **Evaluation**
  - **Optimization**

Representation

- Decision trees
- Sets of rules / Logic programs
- Instances
- Graphical models (Bayes/Markov nets)
- Neural networks
- Support vector machines
- Model ensembles
- Etc.

Evaluation

- Accuracy
- Precision and recall
- Squared error
- Likelihood
- Posterior probability
- Cost / Utility
- Margin
- Entropy
- K-L divergence
- Etc.

Optimization

- Combinatorial optimization
  - E.g.: Greedy search
- Convex optimization
  - E.g.: Gradient descent
- Constrained optimization
  - E.g.: Linear programming

Types of Learning

- **Association Analysis**
- **Supervised (inductive) learning**
  - Training data includes desired outputs
  - Classification
  - Regression/Prediction
- **Unsupervised learning**
  - Training data does not include desired outputs
- **Semi-supervised learning**
  - Training data includes a few desired outputs
- **Reinforcement learning**
  - Rewards from sequence of actions

Learning Associations

- Basket analysis:

  $P(Y \mid X)$ probability that somebody who buys X also buys Y where X and Y are products/services.

  Example: P (Bread | Milk) = 0.6

Market-Basket transactions

TID	*Items*
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Supervised Learning

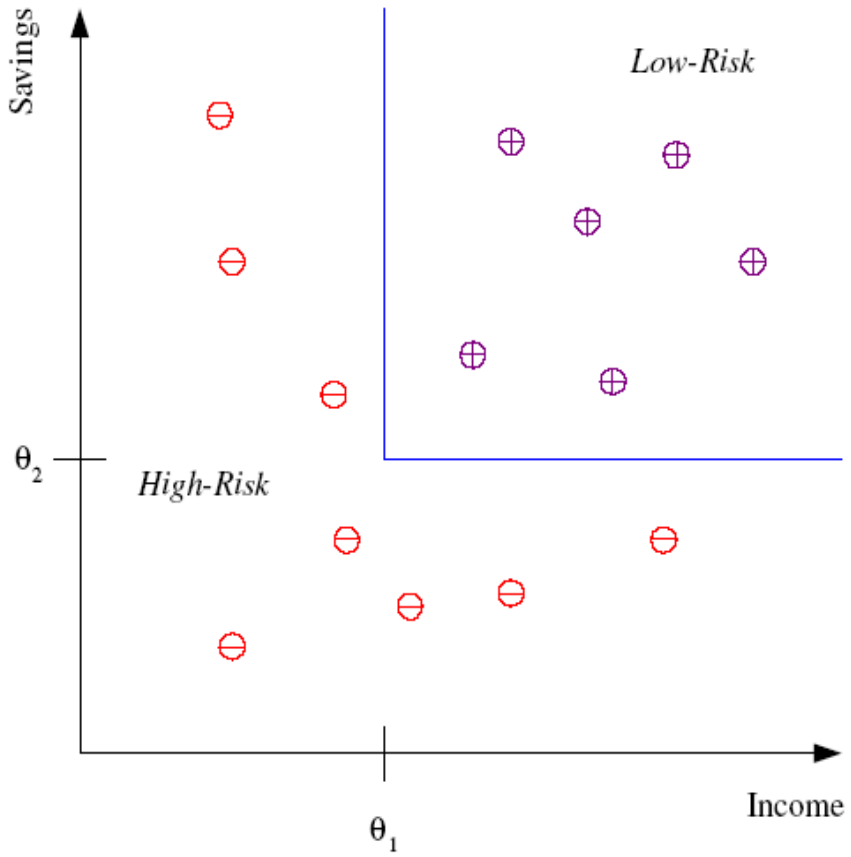
- Prediction of future cases: Use the rule to predict the output for future inputs
- Knowledge extraction: The rule is easy to understand
- Compression: The rule is simpler than the data it explains
- Outlier detection: Exceptions that are not covered by the rule, e.g., fraud

Techniques

- **Supervised learning**
  - Decision tree induction
  - Rule induction
  - Instance-based learning
  - Bayesian learning
  - Neural networks
  - Support vector machines
  - Model ensembles
  - Learning theory

Classification

- Example: Credit scoring
- Differentiating between low-risk and high-risk customers from their *income* and *savings*



Discriminant: IF *income* > θ_1 AND *savings* > θ_2
THEN low-risk ELSE high-risk

Model

Classification: Applications

- Face recognition: Pose, lighting, occlusion (glasses, beard), make-up, hair style
- Character recognition: Different handwriting styles.
- Speech recognition: Temporal dependency.
  - Use of a dictionary or the syntax of the language.
  - Sensor fusion: Combine multiple modalities; eg, visual (lip image) and acoustic for speech
- Medical diagnosis: From symptoms to illnesses
- Web Advertizing: Predict if a user clicks on an ad on the Internet.

Face Recognition

Training examples of a person



Test images



AT&T Laboratories, Cambridge UK
http://www.uk.research.att.com/facedatabase.html

Prediction: Regression

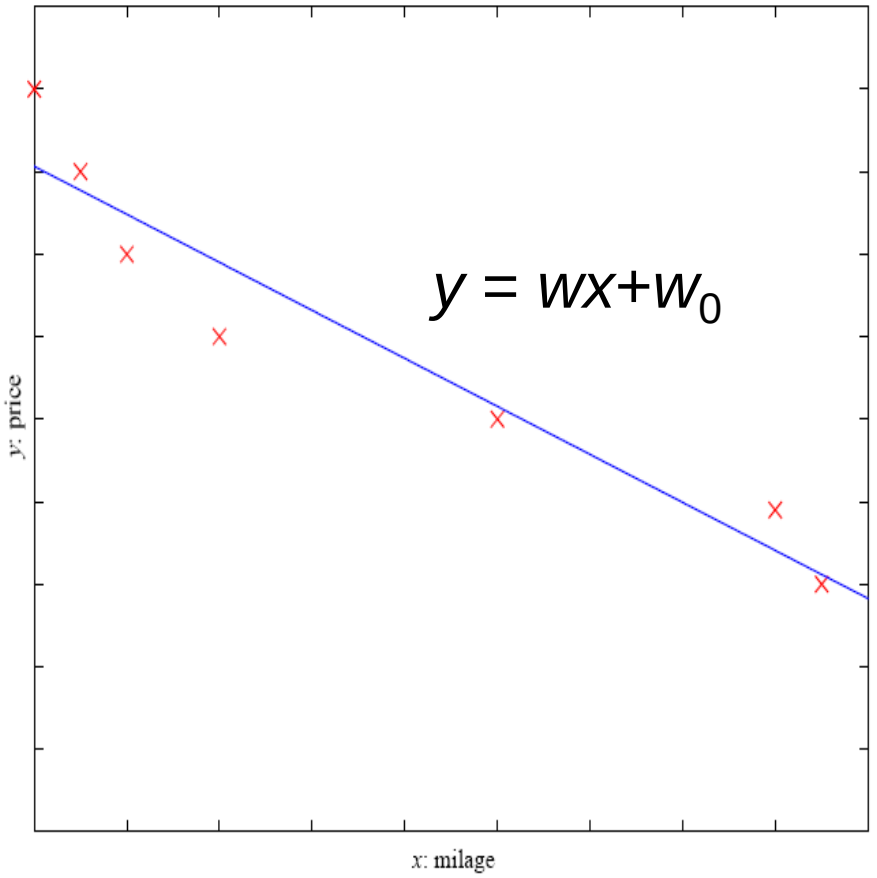
- Example: Price of a used car
- x : car attributes

  y : price

  $$y = g(x \mid \theta)$$

  $g(\ )$ model,

  θ parameters



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Unsupervised Learning

- Learning “what normally happens”
- No output
- Clustering: Grouping similar instances
- Other applications: Summarization, Association Analysis
- Example applications
  - Customer segmentation in CRM
  - Image compression: Color quantization
  - Bioinformatics: Learning motifs

Techniques

- **Unsupervised learning**
  - Clustering
  - Dimensionality reduction

Reinforcement Learning

- Topics:
  - Policies: what actions should an agent take in a particular situation
  - Utility estimation: how good is a state (used by policy)
- No supervised output but delayed reward
- Credit assignment problem (what was responsible for the outcome)
- Applications:
  - Game playing
  - Robot in a maze
  - Multiple agents, partial observability, ...

Solving Real World Problems

- What Is the Input?
  - Features representing the real world data
- What Is the Output?
  - Predictions or decisions to be made
- What Is the Intelligent Program?
  - Types of classifiers, value functions, etc.
- How to Learn from experience?
  - Learning algorithms

Simple illustrative learning problem

Problem:

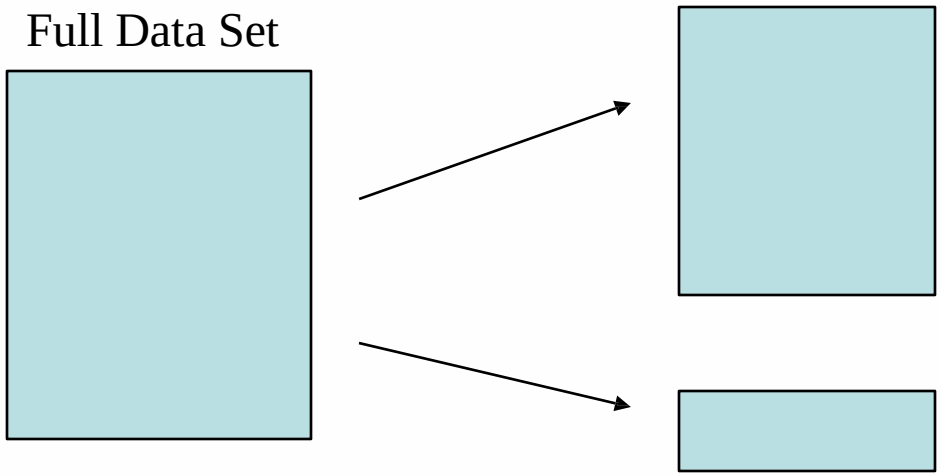
Decide whether to wait for a table at a restaurant, based on the following attributes:

1. Alternate: is there an alternative restaurant nearby?
2. Bar: is there a comfortable bar area to wait in?
3. Fri/Sat: is today Friday or Saturday?
4. Hungry: are we hungry?
5. Patrons: number of people in the restaurant (None, Some, Full)
6. Price: price range ($, $$, $$$)
7. Raining: is it raining outside?
8. Reservation: have we made a reservation?
9. Type: kind of restaurant (French, Italian, Thai, Burger)
10. Wait Estimate: estimated waiting time (0-10, 10-30, 30-60, >60)

Training Data for Supervised Learning

Example	Attributes										Target
	Alt	*Bar*	*Fri*	*Hun*	*Pat*	*Price*	*Rain*	*Res*	*Type*	*Est*	*Wait*
X_1	T	F	F	T	Some	$$$	F	T	French	0–10	T
X_2	T	F	F	T	Full	$	F	F	Thai	30–60	F
X_3	F	T	F	F	Some	$	F	F	Burger	0–10	T
X_4	T	F	T	T	Full	$	F	F	Thai	10–30	T
X_5	T	F	T	F	Full	$$$	F	T	French	>60	F
X_6	F	T	F	T	Some	$$	T	T	Italian	0–10	T
X_7	F	T	F	F	None	$	T	F	Burger	0–10	F
X_8	F	F	F	T	Some	$$	T	T	Thai	0–10	T
X_9	F	T	T	F	Full	$	T	F	Burger	>60	F
X_{10}	T	T	T	T	Full	$$$	F	T	Italian	10–30	F
X_{11}	F	F	F	F	None	$	F	F	Thai	0–10	F
X_{12}	T	T	T	T	Full	$	F	F	Burger	30–60	T

Training and Validation Data



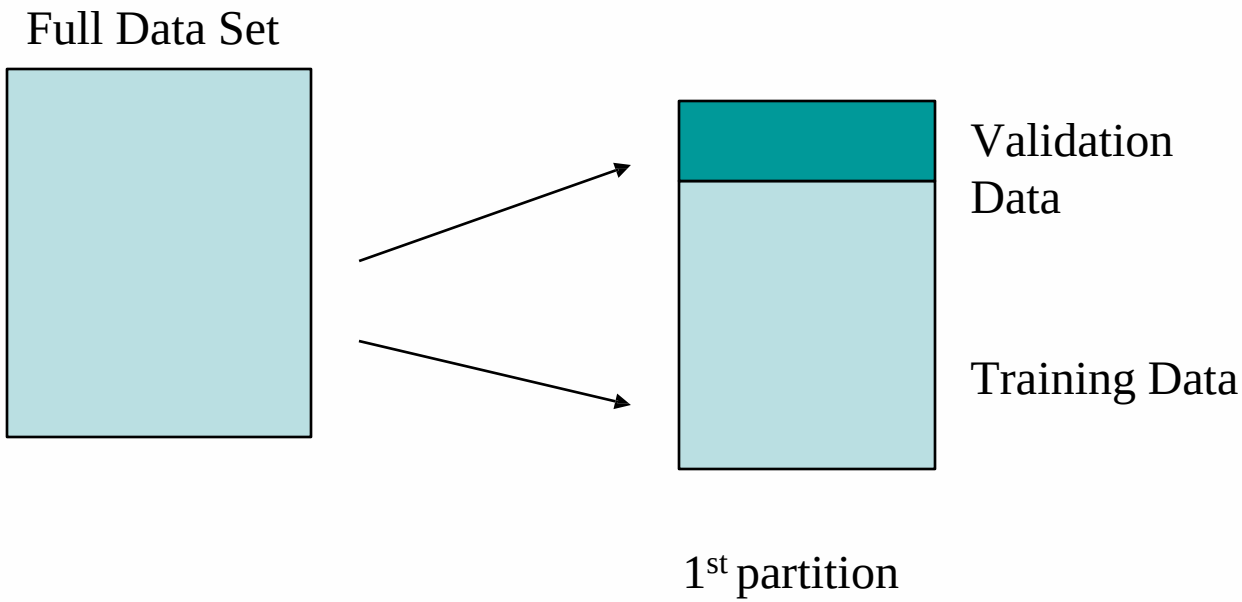
Idea: train each model on the "training data"

and then test each model's accuracy on the validation data

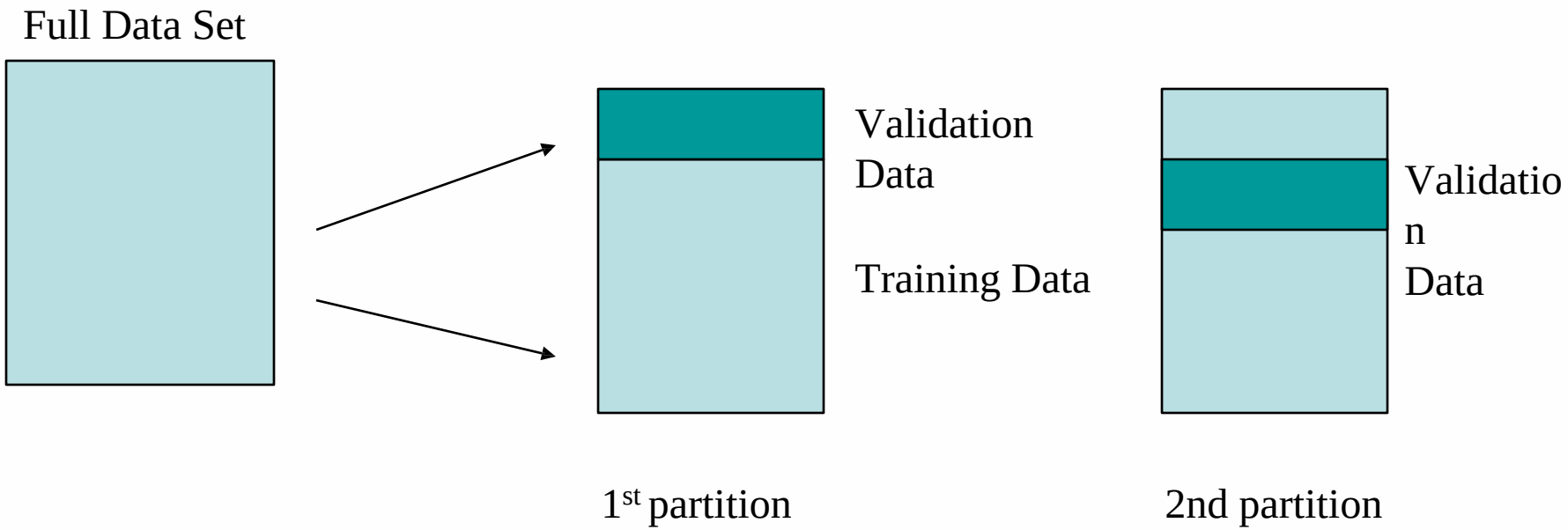
The k-fold Cross-Validation Method

- Why just choose one particular 90/10 “split” of the data?
  - In principle we could do this multiple times
- “k-fold Cross-Validation” (e.g., k=10)
  - randomly partition our full data set into k disjoint subsets (each roughly of size n/k, n = total number of training data points)
    - for i = 1:10 (here k = 10)
      - train on 90% of data,
      - Acc(i) = accuracy on other 10%
    - end
  - common values for k are 5 and 10 (depens on data size)
  - Can also do “leave-one-out” where k = n

Disjoint Validation Data Sets



Disjoint Validation Data Sets



- Notes
  - cross-validation generates an approximate estimate of how well the learned model will do on “unseen” data
  - by averaging over different partitions it is more robust than just a single train/validate partition of the data
  - “k-fold” cross-validation is a generalization
    - partition data into disjoint validation subsets of size n/k
    - train, validate, and average over the k partitions
    - e.g., k=10 is commonly used
  - k-fold cross-validation is approximately k times computationally more expensive than just fitting a model to all of the data

Lab Exercise

Go to UCI Machine Learning Repository
https://archive.ics.uci.edu/ml/index.php

Browse dataset with different type of applications

Download Iris dataset

Conduct data preprocessing – Normalization and k-fold